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United States Patent	12389594
Kind Code	B2
Date of Patent	August 12, 2025
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Integrated circuit device and electronic system including same

Abstract

An integrated circuit device includes; a semiconductor substrate having a cell region and a connection region, a gate stack including gate electrodes and insulating layers alternately stacked and having a stair-stepped structure in the connection region covered by a cover insulating layer. The gate electrodes extend from the cell region to the connection region to terminate in a corresponding pad portion and include lower gate electrodes disposed in a lower portion of the stair-stepped structure, upper gate electrodes disposed in an upper portion of the stair-stepped structure, and middle gate electrodes disposed between the lower gate electrodes and the upper gate electrodes. A contact plug disposed in the connection region includes; first contact plug portions contacting pad portions of the lower gate electrodes, second contact plug portions contacting pad portions of the middle gate electrodes, and third contact plug portions contacting pad portions of the upper gate electrodes. A second diameter of the second contact plug portions is less than a first diameter of the first contact plug portions, and the second diameter is less than a third diameter of the third contact plug portions.

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Appl. No.:	17/549924
Filed:	December 14, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220208781 A1	Jun. 30, 2022

Foreign Application Priority Data

KR

10-2020-0184699

Dec. 28, 2020

Publication Classification

Int. Cl.: H10B41/27 (20230101); G11C5/02 (20060101); H10B43/27 (20230101)

U.S. Cl.:

CPC H10B41/27 (20230201); G11C5/025 (20130101); H10B43/27 (20230201);

Field of Classification Search

CPC: H10B (41/27)

USPC: 257/314

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0184699 filed on Dec. 28, 2020 in the Korean Intellectual Property Office, the subject matter of which is hereby incorporated by reference.

BACKGROUND

(2) The inventive concept relates generally integrated circuit devices and electronic systems

including same. More particularly, the inventive concept relates to integrated circuit devices including a nonvolatile vertical memory device and an electronic system including same.

(3) It is required to increase a degree of integration of an integrated circuit device to provide high performance and realize economic feasibility. In particular, a degree of integration of a memory device is an important factor that determines the economic feasibility of a product. A degree of integration of a two-dimensional memory device is largely determined by an area of a unit memory cell, and is thus greatly affected by the level of micro-pattern formation technology. However, expensive equipment is required to form micro-patterns, and an area of a chip die is limited, and thus, although the degree of integration of a two-dimensional memory device is increasing, the increase is still limited. Thus, a vertical memory device having a three-dimensional structure is required.

SUMMARY

(4) The inventive concept provides integrated circuit devices exhibiting improved product reliability, as well as electronic systems including such integrated circuit devices. Among other beneficial aspects, embodiments of the inventive concept provide improved product reliability by reducing difficulties in the manufacturing process of certain integrated circuit devices. Among other aspects, this may be accomplished by providing a contact plug including multiple contact plug portions have differently sized diameters. These contact plug portions may be used to respectively connect gate electrodes extending into a connection region.

(5) According to an aspect of the inventive concept, there is provided an integrated circuit device including; a semiconductor substrate having a cell region and a connection region adjacent to the cell region, a gate stack including gate electrodes and insulating layers alternately stacked and having a stair-stepped structure in the connection region covered by a cover insulating layer, wherein each one of the gate electrodes extends from the cell region to the connection region to terminate in a corresponding pad portion, and the gate electrodes include lower gate electrodes disposed in a lower portion of the stair-stepped structure, upper gate electrodes disposed in an upper portion of the stair-stepped structure, and middle gate electrodes disposed between the lower gate electrodes and the upper gate electrodes, and a contact plug disposed in the connection region and including; first contact plug portions extending through the cover insulating layer to respectively contact pad portions of the lower gate electrodes, second contact plug portions extending through the cover insulating layer to respectively contact pad portions of the middle gate electrodes, and third contact plug portions extending through the cover insulating layer to respectively contact pad portions of the upper gate electrodes, wherein a second diameter at an upper surface of at least one of the second contact plug portions is less than a first diameter at an upper surface of at least one of the first contact plug portions, and the second diameter is less than a third diameter at an upper surface of at least one of the third contact plug portions.

(6) According to an aspect of the inventive concept, there is provided an integrated circuit device including; a semiconductor substrate having a cell region and a connection region adjacent to the cell region, a gate stack including gate electrodes and insulating layers alternately stacked, wherein each one of the gate electrodes extends from the cell region to the connection region to terminate in a corresponding pad portion, and the gate electrodes include first gate electrodes, second gate electrodes, third gate electrodes and fourth gate electrodes, channel structures disposed in the cell region and extending through the gate stack, and a contact plug disposed in the connection region and including first contact plug portions respectively contacting pad portions of the first gate electrodes, second contact plug portions respectively contacting pad portions of second gate electrodes, third contact plug portions respectively contacting pad portions of the third gate electrodes, and fourth contact plug portions respectively contacting pad portions of the fourth gate electrodes, wherein a second diameter at an upper surface of any one of the second contact plug portions is less than a first diameter at an upper surface of any one of the first contact plug portions.

(7) According to an aspect of the inventive concept, there is provided an electronic system

including a substrate, an integrated circuit device on the substrate, and a controller on the substrate and electrically connected to the integrated circuit device. The integrated circuit device includes; a semiconductor substrate having a cell region and a connection region adjacent to the cell region, a gate stack including gate electrodes and insulating layers alternately stacked and having a stair-stepped structure in the connection region, wherein each one of the gate electrodes extends from the cell region to the connection region to terminate at a corresponding pad portion, and the gate electrodes include lower gate electrodes disposed in a lower portion of the stair-stepped structure, upper gate electrodes disposed in an upper portion of the stair-stepped structure, and middle gate electrodes disposed between the lower gate electrodes and the upper gate electrodes, a contact plug disposed in the connection region and including first contact plug portions respectively contacting pad portions of the lower gate electrodes, second contact plug portions respectively contacting pad portions of the middle gate electrodes, and third contact plug portions respectively contacting pad portions of the upper gate electrodes. A second diameter at an upper surface of at least one of the second contact plug portions is less than a first diameter at an upper surface of at least one of the first contact plug portions, and the second diameter is less than a third diameter at an upper surface of at least one of the third contact plug portions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the inventive concept may be more clearly understood upon consideration of the following detailed description together with the accompanying drawings in which:
- (2) FIG. 1 is a block diagram illustrating an integrated circuit device according to an embodiment of the inventive concept;
- (3) FIG. 2 is an equivalent circuit diagram of a memory cell array of an integrated circuit device according to an embodiment of the inventive concept;
- (4) FIG. 3 is a plan view illustrating components of an integrated circuit device according to an embodiment of the inventive concept;
- (5) FIG. 4 is a cross-sectional view taken along line IV-IV' of FIG. 3;
- (6) FIG. 5 is an enlarged view of region 'V' indicated in FIG. 4; and
- (7) FIG. 6 is an enlarged view of region 'VI' indicated in FIG. 4;
- (8) FIG. 7 is a graph illustrating surface recess of a pad portion in relation to etching depth of a contact hole;
- (9) FIGS. 8, 9, 10, 11 and 12 (collectively, "FIGS. 8 to 12") are respective, cross-sectional views of integrated circuit devices (200, 300, 400 and 500) according to various embodiments of the inventive concept;
- (10) FIGS. 13A, 13B, 13C, 13D and 13E (collectively, "FIGS. 13A to 13E") are related cross-sectional views illustrating a method of manufacturing an integrated circuit device according to an embodiment of the inventive concept;
- (11) FIG. 14 is a conceptual diagram illustrating an electronic system including an integrated circuit device according to an embodiment of the inventive concept;
- (12) FIG. 15 is a perspective view further illustrating an electronic system including an integrated circuit device according to an embodiment of the inventive concept; and
- (13) FIGS. 16 and 17 are respective, cross-sectional views illustrating a semiconductor package including an integrated circuit device according to an embodiment of the inventive concept.

DETAILED DESCRIPTION

- (14) Throughout the written description and drawings, like reference numbers and labels are used to denote like or similar elements and/or features. Throughout the written description certain geometric terms may be used to highlight relative relationships between elements, components

and/or features with respect to certain embodiments of the inventive concept. Those skilled in the art will recognize that such geometric terms are relative in nature, arbitrary in descriptive relationship(s) and/or directed to aspect(s) of the illustrated embodiments. Geometric terms may include, for example: height/width; vertical/horizontal; top/bottom; higher/lower; closer/farther; thicker/thinner; proximate/distant; above/below; under/over; upper/middle/lower; center/side; center/surrounding; overlay/underlay, etc.

(15) FIG. 1 is a block diagram illustrating an integrated circuit device according to an embodiment of the inventive concept.

(16) Referring to FIG. 1, an integrated circuit device **10** may include a memory cell array **20** and a peripheral circuit **30**.

(17) The memory cell array **20** may include memory cell blocks BLK1, BLK2, . . . , BLKn. Each of the memory cell blocks BLK1, BLK2, . . . , BLKn may include a vast multiplicity of memory cells. The memory cell blocks BLK1, BLK2, . . . , BLKn may be connected to the peripheral circuit **30** via a bit line BL, a word line WL, a string select line SSL, and a ground select line GSL.

(18) The memory cell array **20** may be connected to a page buffer **34** via the bit line BL, and connected to a row decoder **32** via the word line WL, the string select line SSL, and the ground select line GSL. In some embodiments, in the memory cell array **20**, each of the memory cells included in the memory cell blocks BLK1, BLK2, . . . , BLKn may be a flash memory cell. The memory cell array **20** may include a three-dimensional (3D) memory cell array. The 3D memory cell array may include NAND strings, wherein each of the NAND strings may include vertically stacked memory cells connected to word lines WL.

(19) The peripheral circuit **30** may include the row decoder **32**, the page buffer **34**, a data I/O circuit **36**, and a control logic **38**. Although not illustrated, the peripheral circuit **30** may further include various circuits such as a voltage generation circuit generating various voltages needed for an operation of the integrated circuit device **10**, an error correction circuit correcting an error of data read from the memory cell array **20**, and an input/output (I/O) interface, or the like.

(20) The peripheral circuit **30** may receive at least one of address(es) ADDR, command(s) CMD, and control signal(s) CTRL from an external source, and may communicate (i.e., transmit and/or receive) data DATA with one or more external source(s).

(21) One exemplary configuration for the peripheral circuit **30** is described hereafter.

(22) The row decoder **32** may select at least one of the memory cell blocks BLK1, BLK2, . . . , BLKn in response to an address ADDR, and may select a word line WL, a string select line SSL, and/or a ground select line GSL of the selected memory cell block. The row decoder **32** may transfer a voltage for performing a memory operation to the word line WL of the selected memory cell block.

(23) The page buffer **34** may be connected to the memory cell array **20** via the bit line BL. The page buffer **34** may operate as a write driver during a write (or program) operation, and may apply to the bit line BL a voltage according data DATA to be stored in the memory cell array **20**. The page buffer **34** may also operate as a sense amplifier during a read operation and sense data DATA stored in the memory cell array **20**. The page buffer **34** may operate according to a control signal PCTL provided from the control logic **38**.

(24) The data I/O circuit **36** may be connected to the page buffer **34** via data line DLs. The data I/O circuit **36** may receive data DATA from a memory controller (not shown) during a program operation, and provide program data DATA to the page buffer **34** in response to a column address C_ADDR provided from the control logic **38**. The data I/O circuit **36** may provide read data DATA stored in the page buffer **34** to the memory controller during a read operation in response to the column address C_ADDR provided from the control logic **38**. The data I/O circuit **36** may transfer the applied address and/or command to the control logic **38** or the row decoder **32**.

(25) The control logic **38** may receive a command CMD and a control signal CTRL from the memory controller. The control logic **38** may provide a row address R_ADDR to the row decoder

32 and a column address C_ADDR to the data I/O circuit **36**. The control logic **38** may generate various internal control signals used in the integrated circuit device **10** in response to the control signal CTRL. For example, when performing a memory operation such as a program operation or an erase operation, the control logic **38** may adjust a voltage level provided to the word line WL and the bit line BL.

(26) FIG. 2 is a partial equivalent circuit diagram of a memory cell array of an integrated circuit device according to an embodiment, and assumes the use of a vertical NAND flash memory device having a vertical channel structure.

(27) A memory cell array MCA may include memory cell strings MS. The memory cell array MCA may include bit lines BL, word lines WL, at least one string select line SSL, at least one ground select line GSL, and a common source line CSL.

(28) The memory cell strings MS may be respectively disposed between the bit lines BL and the common source line CSL. While the memory cell strings MS may include two string select lines SSL as illustrated in FIG. 2, the inventive concept is not limited thereto. For example, each of the memory cell strings MS may include a single string select line SSL.

(29) Each of the memory cell strings MS may include a string select transistor SST, a ground select transistor GST, and memory cell transistors (e.g., MC1, MC2, . . . , MCn-1, MCn). A drain region of the string select transistor SST may be connected to the bit line BL, and a source region of the ground select transistor GST may be connected to the common source line CSL. The common source line CSL may be a region to which source regions of ground select transistors GST are commonly connected.

(30) The string select transistor SST may be connected to the string select line SSL, and the ground select transistor GST may be connected to the ground select line GSL. The memory cell transistors MC1, MC2, . . . , MCn-1, MCn may be respectively connected to the word lines WL.

(31) FIG. 3 is a plan view illustrating components of an integrated circuit device according to an embodiment of the inventive concept; FIG. 4 is a cross-sectional view taken along line IV-IV' of FIG. 3; FIG. 5 is an enlarged view of region 'V' indicated in FIG. 4; and FIG. 6 is an enlarged view of region 'VI' indicated in FIG. 4.

(32) Referring collectively to FIGS. 3, 4, 5, and 6, an integrated circuit device **100** may include a cell array structure CS including a memory cell region MCR and a connection region CON.

(33) In some embodiments, the memory cell region MCR may be a region in which a memory cell array MCA may be formed. Here, the MCA is assumed to include NAND type memory cells having vertical channel structures like those previously described in relation to FIG. 2. The connection region CON may be a region in which various pad portions PAD may be formed. Here, the pad portions PAD may serve as respective electrical connections between various elements or components disposed in the memory cell region MCR and elements or components disposed in a peripheral circuit region (not shown).

(34) A semiconductor substrate **101** may include a semiconductor material, for example, a Group IV semiconductor, a Group III-V compound semiconductor, or a Group II-VI oxide semiconductor. For example, a Group IV semiconductor may include silicon (Si), germanium (Ge), or Si—Ge. The semiconductor substrate **101** may also be provided as a bulk wafer or a wafer on which an epitaxial layer is formed. In other embodiments, the semiconductor substrate **101** may include a silicon on insulator (SOI) substrate or a germanium on insulator (GeOI) substrate.

(35) A gate stack GS may be disposed in a first direction (e.g., an X direction) and a second direction (e.g., a Y direction) extending substantially in parallel with a principal surface (e.g., an upper surface) of the semiconductor substrate **101**, wherein the first direction intersects the second direction. Here, the gate stack GS may include alternately gate electrodes **130** and insulating layers **140** "stacked" one upon the other in a third direction (e.g., a Z direction) substantially perpendicular to the principal surface of the semiconductor substrate **101**.

(36) Also, an upper insulating layer **150** may be arranged on the gate stack GS.

(37) In some embodiments, each one of the gate electrodes **130** may include a buried conductive layer **132** and an insulating liner **134** surrounding an upper surface, a bottom surface, and a side surface of the buried conductive layer **132**. In this regard, the buried conductive layer **132** may include a metal (e.g., tungsten), a metal silicide (e.g., tungsten silicide), doped polysilicon, or a combination thereof. In some embodiments, the insulating liner **134** may include a high-k dielectric material such as an aluminum oxide.

(38) Each one of the gate electrodes **130** may correspond to the ground select line GSL, one of the word lines WL, and the at least one string select line SSL—referencing the memory cell string MS previously described in relation to FIG. 2. For example, a lowermost gate electrode **130** in the gate stack GS may function as the ground select line GSL, an uppermost gate electrode **130** in the gate stack GS may function as the string select line SSL, and other gate electrodes **130** in the gate stack GS may respectively function as a word line WL. Accordingly, a memory cell string MS may include a serially connected arrangement of the ground select transistor GST, the memory cell transistors MC1, MC2, . . . , MCn-1, MCn, and the string select transistor SST.

(39) Word line cuts **170** may extend on the semiconductor substrate **101** in the first direction. The gate stack GS may be disposed between adjacent word line cuts **170**, such that a separation distance between the adjacent word line cuts **170** defines a width of the gate stack GS in the second direction. In some embodiments, the word line cuts **170** may include an insulation spacer **172** and an insulation separation layer **174**. That is, the word line cuts **170** may include an insulating structure. Common source regions CRS may be formed in the semiconductor substrate **101** in relation to each word line cut **170**, wherein each common source region CRS may be an impurity-doped region doped with a high-density of impurities.

(40) Channel structures **160** may extend in the third direction to respectively pass through the gate electrodes **130** from the upper surface of the semiconductor substrate **101** in the memory cell region MCR. The channel structures **160** may be spaced apart in the first direction and the second direction. The channel structures **160** may be arranged (e.g.,) in a zigzag pattern or a staggered pattern.

(41) Each of the channel structures **160** may extend within a channel hole **160H** to pass through the gate stack GS. In some embodiments, each of the channel structures **160** may include a gate insulating layer **162**, a channel layer **164**, a buried insulating layer **166**, and a conductive plug **168**. The gate insulating layer **162** and the channel layer **164** may be sequentially arranged on a side wall of the channel hole **160H**. For example, the gate insulating layer **162** may be arranged conformally on a side wall of the channel hole **160H**, and the channel layer **164** may be conformally arranged on the side wall and a bottom portion of the channel hole **160H**. The buried insulating layer **166** filling a remaining space of the channel hole **160H** may be arranged on the channel layer **164**. The conductive plug **168** contacting the channel layer **164** and blocking an entrance of the channel hole **160H** (e.g., at an uppermost end) may be arranged in an upper portion of the channel hole **160H**. Alternately, the buried insulating layer **166** may be omitted, and the channel layer **164** may be formed in a pillar shape filling the remaining portion of the channel hole **160H**.

(42) The channel structures **160** may respectively contact the semiconductor substrate **101**. In some embodiments, the channel layer **164** may be arranged to contact the upper surface of the semiconductor substrate **101** on the bottom portion of the channel hole **160H**. Alternately, a contact semiconductor layer (not shown) having a certain height on the semiconductor substrate **101** on the bottom portion of the channel hole **160H** may be formed, and the channel layer **164** may be electrically connected to the semiconductor substrate **101** via the contact semiconductor layer.

(43) The gate insulating layer **162** may include a tunneling dielectric layer **162A**, a charge storage layer **162B**, and a blocking dielectric layer **162C** that are sequentially formed on an outer side wall of the channel layer **164**. Relative thicknesses of the tunneling dielectric layer **162A**, the charge storage layer **162B**, and the blocking dielectric layer **162C** of the gate insulating layer **162** are not

limited to the illustrated ones, and may be modified in various manners.

(44) The tunneling dielectric layer **162A** may include silicon oxide, hafnium oxide, aluminum oxide, zirconium oxide, tantalum oxide, or the like. The charge storage layer **162B** may be a region in which electrons that have passed through the tunneling dielectric layer **162A** from the channel layer **164** may be stored, and may include silicon nitride, boron nitride, silicon boron nitride, or impurity-doped polysilicon. The blocking dielectric layer **162C** may include silicon oxide, silicon nitride, or a metal oxide having a higher permittivity than silicon oxide.

(45) In each memory cell block, an uppermost gate electrode **130** may be divided into two portions two-dimensionally via a string separation insulating layer (not shown). Thus, these two portions may constitute the string select line SSL as previously described in relation to FIG. 2.

(46) In the connection region CON, each of the gate electrodes **130** may (laterally) extend to terminate (or form) the pad portion PAD at a distal end of the gate electrode **130** with respect to the memory cell region MCR, wherein a cover insulating layer **120** may be disposed to cover the pad portion PAD. In some embodiments, the distal end of the gate electrode **130** may have a greater thickness than other portions of the gate electrode **130**. That is, the pad portion PAD may have an upper surface that is “higher” (e.g., with respect to the principal surface of the semiconductor substrate **101**) than the upper surface of other portions of the gate electrode **130**, but the inventive concept is not limited thereto.

(47) As illustrated in FIG. 4, for example, in the connection region CON, each of the gate electrodes **130** stacked between a lowermost gate electrode to an uppermost gate electrode may extend a lesser distance in the first direction than an adjacent, underlying one of the gate electrodes **130**. Hence, the lowermost gate electrode may extend a greatest distance in the first direction, and the uppermost gate electrode **130** may extend a least distance in the first direction. Hereafter, this type of gate electrode arrangement for a gate stack GS in the connection region CON will be referred to as a “stair-stepped structure.”

(48) A contact plug CNT, including multiple contact plug portions, may pass through the cover insulating layer **120** to connect at least one pad portion PAD associated with at least one gate electrode **130**. Thus, each contact plug portion may be arranged in a corresponding contact hole CNTH in the connection region CON. In some embodiments, each contact plug portion may have a tapered pillar shape. In this regard, the term “tapered pillar shape” denotes a shape wherein the diameter (or width, measured in at least one of the first direction or the second direction) of a vertically extending pillar structure decreases as the pillar structure descends in the third direction towards the semiconductor substrate **101**.

(49) In the illustrated example of FIG. 4, the contact plug CNT includes at least first contact plugs CNT1, second contact plugs CNT2, and third contact plugs CNT3, wherein the first contact plugs CNT1 vertically extend to contact lower gate electrode(s) in the stair-stepped structure, second contact plugs CNT2 vertically extend to contact middle gate electrode(s) in the stair-stepped structure, and third contact plugs CNT3 vertically extend to contact upper gate electrode(s) in the stair-stepped structure. As a result, a first distance in the first direction between the memory cell region MCR and any one of the first contact plugs CNT1 will be greater than a second distance between the memory cell region MCR and any one of the second contact plugs CNT2. Further, the second distance will be greater than a third distance between the memory cell region MCR and any one of the third contact plugs CNT3.

(50) In some embodiments, the second contact plug portions respectively contact pad portions of gate electrodes disposed between $\frac{1}{4}$ to $\frac{1}{2}$ of a height of the gate structure as measured from an upper surface of the semiconductor substrate.

(51) In some embodiments, the various contact holes CNTH may be configured such that uppermost surfaces of the different contact plug portions have different diameters that vary with positioning of the contact plug portions within the connection region CON. For example, uppermost surfaces of the first contact plugs CNT1 may have a first diameter D1, uppermost

surfaces of the second contact plugs CNT2 may have a second diameter D2, and uppermost surfaces of the third contact plugs CNT3 may have a third diameter D3, wherein the second diameter D2 may be less than the first and third diameters D1 and D3, and the first diameter D1 may be greater than the third diameter D3. In some embodiments, for example, the second diameter D2 may range from about 90% to about 95% of the first diameter D1, and the second diameter D2 may range from about 95% to about 99% of the third diameter D3. However, these are just illustrative examples and the inventive concept is not limited thereto.

(52) Alternately, respective diameters for a particular type (or group) of contact plug portions (e.g., the first contact plug portions CNT1) may vary with layout configuration and/or connection purpose. For example, a first diameter of one of the first contact plug portions CNT1 contacting the lowermost gate electrode among the gate electrodes 130 may be greater than a diameter of another one of the first contact plug portions CNT1.

(53) However, in some embodiments, respective first diameters D1 of first contact plugs CNT1 may be substantially the same, respective second diameters D2 of the second contact plugs CNT2 may be substantially the same, and/or respective third diameters D3 of third contact plugs CNT3 may be substantially the same.

(54) Although not specifically illustrated in FIG. 4, one of more dummy channel structures may extend through the gate stack GS in the third direction in the connection region CON. The dummy channel structures may be formed to prevent curving, skewing or bending of the gate stack GS during manufacture of the integrated circuit device 100 to better ensure structural stability.

(55) A bit line contact BLC may pass through the upper insulating layer 150 to electrically connect the conductive plug 168 of the channel structures 160 with the bit line BL extending across the upper insulating layer 150 in the second direction.

(56) A conductive line ML may be formed on the upper insulating layer 150 in the connection region CON. Although not specifically illustrated in FIG. 4, an upper support layer may be formed between the upper insulating layer 150 and the bit line BL, as well as between the upper insulating layer 150 and the conductive line ML.

(57) As the vertical height of the gate stack GS increases, so too does the aspect ratio of the contact plug CNT. That is, a ratio between a height of the contact plug CNT and a width of the contact plug CNT increases. In particular, assuming a gate structure GS including multiple gate electrodes 130, an aspect ratio for the contact plug CNT may increase.

(58) By way of comparison with the foregoing, in conventional integrated circuit devices, contact holes are configured such that contact plugs arranged in a connection region have an equal diameter, regardless of positioning of respective contact plugs. However, as the number of gate electrodes in gate stacks of conventional integrated circuits increases, contact hole depths for contact plugs associated with the lowermost gate electrode have also increased. These very deep contact holes may result in over-etching of some the contact holes, thereby causing excessive surface recesses in a pad portion of a gate electrode, or causing a contact hole to pass through the pad portion. Thus, as the thickness of a gate electrode portion associated with the pad portion increases to prevent these problems, another problem may arise of increasing the level of difficulty of a replacement process of forming a gate electrode by substituting a sacrificial layer with an electrode layer.

(59) Thus, to address at least the foregoing problems, in the integrated circuit device 100 of FIGS. 3, 4, 5, and 6, various contact holes CNTH may be configured with different diameters with respect to two or more contact plug portions (e.g., CNT1, CNT2, and CNT3) that respectively contact the pad portions PAD associated with gate electrodes 130 in the connection region CON.

(60) In this regard, an examination of surface recesses of the pad portion PAD, as a function of etching depth for a contact hole CNTH have been conducted. (See, e.g., FIG. 7). Accordingly, the diameter of some contact holes (e.g., those associated with the second contact plugs CNT2 contacting the middle layers of the stair-stepped structure) should be different from the diameter of

other contact holes (e.g., those associated with structure and the first contact plugs CNT1 and/or the third contact plugs CNT3 respectively contacting the lower layers and upper layers of the stair-stepped structure).

(61) That is, embodiments of the inventive concept greatly reduce or eliminate the possibility of contact hole over-etching, while still providing pad portions of substantially uniform thickness. For example, with respect to the integrated circuit device **100** of FIGS. **3**, **4**, **5** and **6**, there is an ultimate effect of reducing the level of difficulty of a manufacturing process and increasing product reliability, as defects caused by contact hole etching are effectively reduced by appropriately providing differently sized contact holes.

(62) FIG. **7** is a graph illustrating surface recess (SR) of a pad portion as a function of contact hole etching depth (ED).

(63) Referring to FIGS. **4** and **7**, until contact hole etching depth ED reaches a critical depth ED1, the greater the contact hole etching depth ED, the greater also the surface recess SR of the pad portion PAD. However, from the point in time at which the contact hole etching depth ED exceeds the critical depth ED1, the surface recess SR of the pad portion PAD continuously decreases as the contact hole etching depth ED increases.

(64) That is, contact hole etching depth ED less than the critical depth ED1 may correspond to contact holes associated with upper layer(s) of the stair-stepped structure. Hence, as the contact hole etching depth ED is relatively small, the surface recess SR of the pad portion PAD may also be relatively small.

(65) However, contact hole etching depth ED greater than the critical depth ED1 may correspond to contact holes associated with lower layer(s) of the stair-stepped structure. Hence, although the contact hole etching depth ED is relatively great, etching byproducts attach to the contact hole as etching is performed, there resulting in surface recess SR of the pad portion PAD that is also small.

(66) Further, contact hole etching depth ED similar to (or about the same as) the critical depth ED1 may correspond to contact holes associated with middle layer(s) of the stair-stepped structure. In this case, while the contact hole etching depth ED may be relatively middling in depth, the surface recess SR may nonetheless be proportional to the contact hole etching depth ED. Hence, the surface recess SR of the pad portion PAD may be the greatest in relation to contact hole etching depths ED similar to the critical depth.

(67) Thus, based on the foregoing, in order to compensate for a difference in the surface recess SR of the pad portion PAD in integrated circuit devices with respect to embodiments of the inventive concept, the size (or diameter) of contact hole(s) associated with middle layer(s) should be less than the size of contact holes associated with lower layer(s) and/or the size of contact holes associated with upper layer(s).

(68) FIGS. **8** to **12** are cross-sectional views of integrated circuit devices (**200**, **300**, **400** and **500**) according to various embodiments of the inventive concept.

(69) Most of the elements, components and/or features of integrated circuit devices **200**, **300**, **400**, and **500** are substantially similar to those described previously described in relation to FIGS. **3**, **4**, **5**, and **6**. Thus, for convenience of description, only material differences will be highlighted between the integrated circuit device **100** and the integrated circuit devices **200**, **300**, **400** and **500**. Here, FIG. **10** is an enlarged view of region 'X' indicated in FIG. **9**.

(70) Referring to FIG. **8**, the integrated circuit device **200** may further include, with respect to the integrated circuit device **100** of FIG. **4**, a peripheral circuit structure PS, as well as a cell array structure CS disposed at a higher level than the peripheral circuit structure PS with respect to the semiconductor substrate **101**.

(71) The integrated circuit device **200** may have a cell on periphery (COP) structure in which the cell array structure CS is arranged on the peripheral circuit structure PS. A base structure **110** may be arranged between the peripheral circuit structure PS and the cell array structure CS.

(72) The peripheral circuit structure PS may include a peripheral circuit transistor **60TR** and a

peripheral circuit wiring **70** arranged on the semiconductor substrate **101**. An active region AC may be defined in the semiconductor substrate **101** by using a device isolation layer **102**, and peripheral circuit transistors **60TR** may be formed on the active region AC. The peripheral circuit transistors **60TR** may include a peripheral circuit gate **60G** and a source/drain region **62** arranged in a portion of the semiconductor substrate **101** on two sides of the peripheral circuit gate **60G**.

(73) The peripheral circuit wiring **70** may include peripheral circuit contacts **72** and peripheral circuit metal layers **74**. An interlayer insulating layer **80** covering the peripheral circuit transistors **60TR** and the peripheral circuit wiring **70** may be arranged on the semiconductor substrate **101**. The peripheral circuit metal layers **74** may have a multi-layer structure including metal layers arranged at different vertical levels. Although the peripheral circuit metal layers **74** are illustrated to be at a same height in FIG. **8**, alternatively, a peripheral circuit metal layer **74** arranged at some levels (for example, arranged at an uppermost level) may be formed at a higher height than the other peripheral circuit metal layers **74** arranged at other levels.

(74) Here, the second diameter D2 (measured at upper ends) of the second contact plugs CNT2 may be less than the first and third diameters D1 and D3 of the first and third contact plugs CNT1 and CNT3. Also, the first diameter D1 may be greater than the third diameter D3.

(75) Referring to FIGS. **9** and **10**, the integrated circuit device **300** may include a first gate stack GS1 and a second gate stack GS2 over the first gate stack GS1.

(76) In the integrated circuit device **300** of FIGS. **9** and **10**, the first gate stack GS1 may include first gate electrodes **130** and first insulating layers **140** alternately stacked in the third direction. Also, a first upper insulating layer may be arranged on an uppermost portion of the first gate stack GS1.

(77) The second gate stack GS2 may include second gate electrodes **230** and second insulating layers **240**, alternately stacked in the third direction. Also, a second upper insulating layer **250** may be arranged on an uppermost portion of the second gate stack GS2. In the connection region CON, the first gate stack GS1 may have a first stair-stepped structure, and the second gate stack GS2 may have a second stair-stepped structure.

(78) The channel structures **160** may extend in a first channel hole **160H1** passing through the first gate stack GS1 and a second channel hole **160H2** passing through the second gate stack GS2. The channel structures **160** may have a shape that outwardly protrudes at a boundary between the first channel hole **160H1** and the second channel hole **160H2**.

(79) The channel structures **160** may pass through the base structure **110** including an upper base layer **110U** and a lower base layer **110L** to contact the semiconductor substrate **101**. The gate insulating layer **162** may be removed at a same level as the lower base layer **110L** and the channel layer **164** may contact an extension portion **110LE** of the lower base layer **110L**. A side wall portion **162S** and a bottom portion **162L** of the gate insulating layer **162** may be spaced apart with the extension portion **110LE** of the lower base layer **110L** therebetween, and the bottom portion **162L** of the gate insulating layer **162** may be arranged to surround a bottom surface of the channel layer **164**. Accordingly, the channel layer **164** may be electrically connected to the lower base layer **110L**, instead of directly contacting the semiconductor substrate **101**.

(80) In the illustrated embodiment of FIGS. **9** and **10**, the contact plug CNT may include first through fourth contact plug portions CNT1, CNT2, CNT3, and CNT4. That is, the first through fourth contact plug portions CNT1, CNT2, CNT3, and CNT4 may include first contact plug portions CNT1 vertically extending to contact gate electrodes arranged in a lower portion of the first stair-stepped structure, second contact plug portions CNT2 vertically extending to contact gate electrodes arranged in an upper portion of the first stair-stepped structure, third contact plug portions CNT3 vertically extending to contact gate electrodes arranged in a lower portion of the second stair-stepped structure, and fourth contact plug portions CNT4 vertically extending to contact gate electrodes arranged in an upper portion of the second stair-stepped structure.

(81) Here, the second diameter D2 for the second contact plug portions CNT2 may be less than the

first, third, and fourth diameters **D1**, **D3**, and **D4** for the first, third, and fourth contact plug portions **CNT1**, **CNT3**, and **CNT4**. Also, the first diameter **D1** may be greater than the third and fourth diameters **D3** and **D4**.

(82) Referring to FIG. **11**, the integrated circuit device **400** may include the peripheral circuit structure **PS** of FIG. **8**, and the first gate stack **GS1** and the second gate stack **GS2** of FIG. **9**. Hence, the integrated circuit device **400** has features of both the integrated circuit device **200** and the integrated circuit device **300**.

(83) Referring to FIG. **12**, the integrated circuit device **500** may have a chip-to-chip structure. That is, the integrated circuit device **500** has a chip-to-chip structure which may be formed by manufacturing an upper chip including a cell array structure **CS** and a lower chip including a peripheral circuit structure **PS** and then connecting the upper chip and the lower chip to each other by using a bonding method.

(84) Here, in some embodiments, the bonding method may refer to a method in which a bonding pad formed in an uppermost portion of an upper chip is brought into contact with a bonding pad formed in an uppermost portion of a lower chip. The bonding method may include a metal-metal bonding structure, a through silicon via (TSV), a back via stack (BVS), a eutectic bonding structure, a ball grid array (BGA) bonding structure, wiring lines, or a combination thereof.

(85) The peripheral circuit structure **PS** may include a circuit board **301**, an interlayer insulating layer **310**, circuit elements **360**, a first metal layer **330** connected to each of the circuit elements **360**, and a second metal layer **340** formed on the first metal layer **330**.

(86) The interlayer insulating layer **310** may be arranged on the circuit board **301** to cover the circuit elements **360**, the first metal layer **330**, and the second metal layer **340**, and include an insulating material.

(87) A lower bonding pad **370** may be formed on the second metal layer **340** in a word line bonding area **BA1**. In the word line bonding area **BA1**, the lower bonding pad **370** of the peripheral circuit structure **PS** may be electrically connected to an upper bonding pad **470** of the cell array structure **CS** by using a bonding method.

(88) The cell array structure **CS** may provide at least one memory block. The cell array structure **CS** may include a cell substrate **401** and a common source line **CSL**. Word lines **430** may be stacked on the cell substrate **401** in the third direction.

(89) In a bit line bonding area **BA2**, a channel structure **460** may pass through the word lines **430**, string select lines, and a ground select line in the third direction.

(90) In the word line bonding area **BA1**, the word lines **430** may extend in parallel to an upper surface of the cell substrate **401** and be connected to the contact plugs **CNT**. The word lines **430** and the contact plugs **CNT** may be connected in the pad portion **PAD** provided as at least some of the word lines **430** extend at different lengths.

(91) In an external pad bonding area **PA**, a common source line contact **480** may be arranged. The common source line contact **480** may include a conductive material such as a metal, a metal compound, or polysilicon, and may be electrically connected to the common source line **CSL**.

(92) In the external pad bonding area **PA**, first and second I/O pads **350** and **450** may be arranged. A lower layer **320** covering a lower surface of the circuit board **301** may be formed under the circuit board **301**, and the first I/O pad **350** may be formed on the lower layer **320**. An upper layer **420** covering the upper surface of the cell substrate **401** may be formed on the cell substrate **401**, and the second I/O pad **450** may be arranged on the upper layer **420**.

(93) Here, the second diameter **D2** of the second contact plugs **CNT2** may be less than the first and third diameters **D1** and **D3** of the first and third contact plugs **CNT1** and **CNT3**. Also, the first diameter **D1** may be greater than the third diameter **D3**.

(94) FIGS. **13A** to **13E** are related cross-sectional views illustrating a method of manufacturing an integrated circuit device according to embodiments of the inventive concept.

(95) Referring to FIG. **13A**, a sacrificial gate stack **SGS** may be formed on a semiconductor

substrate **101**.

(96) The sacrificial gate stack SGS may be formed by alternately layering insulating layers **140** and mold layers **130M** on a principal surface of the semiconductor substrate **101**.

(97) In some embodiments, the insulating layers **140** may include an insulating material such as silicon oxide, silicon oxynitride, or the like. The mold layers **130M** may include a different material from that of the insulating layers **140**, and the mold layers **130M** may include silicon nitride, silicon oxynitride, or impurity-doped polysilicon.

(98) Referring to FIG. **13B**, in the connection region CON, a sacrificial pad portion SPAD may be formed by sequentially patterning the sacrificial gate stack SGS.

(99) The sacrificial gate stack SGS may include a memory cell region MCR and a connection region CON, and may include the sacrificial pad portion SPAD at an end portion of the mold layers **130M** in the connection region CON. Also, the sacrificial pad portion SPAD may have an increased thickness through an additional process. Accordingly, the sacrificial pad portion SPAD may have an upper surface higher than other portions of the mold layer **130M**.

(100) Also, the cover insulating layer **120** covering the sacrificial pad portion SPAD may be formed. The cover insulating layer **120** may include an insulating material such as silicon oxide, silicon nitride, silicon oxynitride, or the like.

(101) Referring to FIG. **13C**, the sacrificial gate stack SGS may be further patterned to form a channel hole **160H**, and a channel structure **160** including, on an inner wall of the channel hole **160H**, a gate insulating layer **162**, a channel layer **164**, a buried insulating layer **166**, and a conductive plug **168** may be formed.

(102) A word line cut opening WLH may be formed by removing a portion of the sacrificial gate stack SGS. The mold layers **130M** exposed on a sidewall of the word line cut opening WLH may be removed, and a conductive material may be buried in portions from which the mold layers **130M** are removed, to form the gate electrodes **130**. That is, a replacement process of forming the gate electrodes **130** by substituting a sacrificial layer with an electrode layer may be performed. Accordingly, the sacrificial gate stack SGS may be the gate stack GS, and the sacrificial pad portion SPAD may be the pad portion PAD.

(103) Referring to FIG. **13D**, a word line cut **170** may be formed by filling an insulating material in the word line cut opening WLH.

(104) The upper insulating layer **150** may be formed on the gate stack GS, and a contact hole CNTH passing through the upper insulating layer **150** and the cover insulating layer **120** to expose an upper surface of the pad portion PAD may be formed.

(105) In some embodiments, the contact hole CNTH may be configured such that uppermost surfaces of the contact plug portions have different diameters depending on positioning of the contact plug portions. (See, e.g., FIG. **13E**).

(106) A bit line contact BLC passing through the upper insulating layer **150** to be electrically connected to the channel structure **160** may be formed.

(107) Referring to FIG. **13E**, the contact hole CNTH may be filled with a conductive material to form the contact plugs CNT electrically connected to the pad portion PAD.

(108) Here, the uppermost surfaces of the contact plug portion CNT1, CNT2, CNT3 have different diameters depending on positioning and respective connection purposes. That is, uppermost surfaces of the first contact plug portions CNT1 may have a first diameter D1, uppermost surfaces of the second contact plug portions CNT2 may have a second diameter D2, and uppermost surfaces of the third contact plug portions CNT3 may have a third diameter D3.

(109) Referring to FIG. **4**, the bit line BL electrically connected to the bit line contact BLC may be formed on the upper insulating layer **150**, and the conductive line ML electrically connected to the contact plug CNT may be formed on the upper insulating layer **150**. That is, by performing the above-described operations, the integrated circuit device **100** may be manufactured.

(110) FIG. **14** is a conceptual diagram illustrating an electronic system **1000** including an integrated

circuit device according to embodiments of the inventive concept.

(111) Referring to FIG. 14, the electronic system **1000** may include an integrated circuit device **1100** and a controller **1200** electrically connected to the integrated circuit device **1100**.

(112) The electronic system **1000** may include a storage device including one or more integrated circuit devices **1100** or an electronic device including the storage device. For example, the electronic system **1000** may include a solid state drive (SSD) device, a universal serial bus (USB) device, a computing device, a medical equipment, or a communication device including at least one integrated circuit device **1100**.

(113) The integrated circuit device **1100** may include a nonvolatile vertical memory device. For example, the integrated circuit device **1100** may include a NAND flash memory device including at least one of the integrated circuit devices **100**, **200**, **300**, **400**, and **500** described above with reference to FIGS. 3 through 12. The integrated circuit device **1100** may include a first structure **1100F** and a second structure **1100S** on the first structure **1100F**. In some embodiments, the first structure **1100F** may be arranged next to the second structure **1100S**.

(114) The first structure **1100F** may include a peripheral circuit structure including a decoder circuit **1110**, a page buffer **1120**, and a logic circuit **1130**. The second structure **1100S** may include a memory cell structure including bit lines BL, a common source line CSL, word lines WL, first and second gate upper lines UL1 and UL2, first and second gate lower lines LL1 and LL2, and memory cell strings CSTR between the bit line BL and the common source line CSL.

(115) In the second structure **1100S**, the memory cell strings CSTR may each include lower transistors LT1 and LT2 that are adjacent to the common source line CSL, upper transistors UT1 and UT2 that are adjacent to the bit line BL, and memory cell transistors MCT arranged between the lower transistors LT1 and LT2 and the upper transistors UT1 and UT2. The number of lower transistors LT1 and LT2 and the number of the upper transistors UT1 and UT2 may be modified in various manners according to embodiments.

(116) In some embodiments, the upper transistors UT1 and UT2 may include a string select transistor, and the lower transistors LT1 and LT2 may include a ground select transistor. The gate lower lines LL1 and LL2 may be respectively gate electrodes of the lower transistors LT1 and LT2. The word lines WL may be gate electrodes of the memory cell transistors MCT, and the gate upper lines UL1 and UL2 may be gate electrodes of the upper transistors UT1 and UT2.

(117) The common source line CSL, the gate lower lines LL1 and LL2, the word lines WL, and the gate upper lines UL1 and UL2 may be electrically connected to the decoder circuit **1110** via first connection wirings **1115** extending to the second structure **1100S** in the first structure **1100F**. The bit lines BL may be electrically connected to the page buffer **1120** via second connection wirings **1125** extending to the second structure **1100S** in the first structure **1100F**.

(118) In the first structure **1100F**, the decoder circuit **1110** and the page buffer **1120** may perform a control operation on at least one of the memory cell transistors MCT. The decoder circuit **1110** and the page buffer **1120** may be controlled by the logic circuit **1130**.

(119) The integrated circuit device **1100** may communicate with the controller **1200** via an I/O pad **1101** electrically connected to the logic circuit **1130**. The I/O pad **1101** may be electrically connected to the logic circuit **1130** via an I/O connection wiring **1135** extending to the second structure **1100S** in the first structure **1100F**.

(120) The controller **1200** may include a processor **1210**, a NAND controller **1220**, and a host interface **1230**. In some embodiments, the electronic system **1000** may include integrated circuit devices **1100**, and in this case, the controller **1200** may control the integrated circuit devices **1100**.

(121) The processor **1210** may control the overall operation of the electronic system **1000** including the controller **1200**. The processor **1210** may operate according to certain firmware, and may control the NAND controller **1220** to access the integrated circuit devices **1100**. The NAND controller **1220** may include a NAND interface **1221** processing communication with the integrated circuit devices **1100**. A control command for controlling the integrated circuit devices **1100**, data to

be stored in memory cell transistors MCT of the integrated circuit devices **1100**, data to be read from the memory cell transistors MCT of the integrated circuit devices **1100**, or the like may be transmitted through the NAND interface **1221**. The host interface **1230** may provide a communication function between the electronic system **1000** and an external host. When receiving a control command from an external host via the host interface **1230**, the processor **1210** may control the integrated circuit devices **1100** in response to the control command.

(122) FIG. **15** is a perspective view of an electronic system including an integrated circuit device according to an embodiment.

(123) Referring to FIG. **15**, an electronic system **2000** according to the embodiment of the inventive concept may include a principal substrate **2001**, a controller **2002** mounted on the principal substrate **2001**, at least one semiconductor package **2003**, and a dynamic random access memory (DRAM) **2004**.

(124) The principal substrate **2001** may include a connector **2006** including pins coupled to an external host. In the connector **2006**, the number and arrangement of the pins may vary according to a communication interface between the electronic system **2000** and the external host. In some embodiments, the electronic system **2000** may communicate with an external host according to one of interfaces such as USB, Peripheral Component Interconnect Express (PCI-Express), Serial Advanced Technology Attachment (SATA), or M-Phy for Universal Flash Storage (UFS). In some embodiments, the electronic system **2000** may operate by power supplied from an external host via the connector **2006**. The electronic system **2000** may further include a power management integrated circuit (PMIC) that distributes power received from the external host to the at least one semiconductor package **2003**. The at least one semiconductor package **2003** and the DRAM **2004** may be connected to the controller **2002** via wiring patterns **2005** formed on the principal substrate **2001**.

(125) The controller **2002** may record data to the at least one semiconductor package **2003**, read data from the at least one semiconductor package **2003**, or improve an operating rate of the electronic system **2000**.

(126) The DRAM **2004** may include a buffer memory to mitigate a difference in speeds of the at least one semiconductor package **2003**, which is a data storage space, and an external host. The DRAM **2004** included in the electronic system **2000** may operate as a type of cache memory, and provide space to temporarily store data in a control operation on the at least one semiconductor package **2003**. When the DRAM **2004** is included in the electronic system **2000**, the controller **2002** may further include a DRAM controller to control the DRAM **2004**, in addition to a NAND controller for controlling the at least one semiconductor package **2003**.

(127) The at least one semiconductor package **2003** may include first and second semiconductor packages **2003a** and **2003b**. Each of the first and second semiconductor packages **2003a** and **2003b** may include a semiconductor package including semiconductor chips **2200**. Each of the first and second semiconductor packages **2003a** and **2003b** may include a package substrate **2100**, semiconductor chips **2200** on the package substrate **2100**, an adhesive layer **2300** arranged on a lower surface of each of the semiconductor chips **2200**, a connection structure **2400** electrically connecting the semiconductor chips **2200** to the package substrate **2100**, and a molding layer **2500** covering the semiconductor chips **2200** and the connection structure **2400** on the package substrate **2100**.

(128) The package substrate **2100** may be a printed circuit board including package upper pads **2130**. Each of the semiconductor chips **2200** may include an I/O pad **2201**. The I/O pad **2201** may correspond to the I/O pad **1101** of FIG. **14**. Each of the semiconductor chips **2200** may include gate stacks **3210** and channel structures **3220**. For example, the semiconductor chips **2200** may include at least one of the integrated circuit devices **100**, **200**, **300**, **400**, and **500** described above with reference to FIGS. **3** through **12**.

(129) In some embodiments, the connection structure **2400** may include a bonding wire electrically

connecting the I/O pad **2201** to the package upper pads **2130**. Accordingly, in the first and second semiconductor packages **2003a** and **2003b**, the semiconductor chips **2200** may be electrically connected to each other by using a bonding wire method, and may be electrically connected to the package upper pads **2130** of the package substrate **2100**. In some embodiments, in the first and second semiconductor packages **2003a** and **2003b**, the semiconductor chips **2200** may be electrically connected to each other via a connection structure including a TSV, instead of the connection structure **2400** using a bonding wire method.

(130) In some embodiments, the controller **2002** and the semiconductor chips **2200** may be included in a single package. In some embodiments, the controller **2002** and the semiconductor chips **2200** may be mounted on the principal substrate **2001** and another, additional interposer substrate, and the controller **2002** and the semiconductor chips **2200** may be connected to each other via a wiring formed on the interposed substrate.

(131) FIGS. **16** and **17** are cross-sectional views illustrating a semiconductor package including an integrated circuit device according to an embodiment.

(132) In FIGS. **16** and **17**, a configuration of a cross-sectional view taken along line A-A' of FIG. **15** is illustrated in detail.

(133) Referring to FIG. **16**, in a semiconductor package **3003**, the package substrate **2100** may be a printed circuit board.

(134) The package substrate **2100** may include a body portion **2120**, package upper pads **2130** (see FIG. **15**) arranged on an upper surface of the body portion **2120**, lower pads **2125** arranged on a lower surface of the body portion **2120** or exposed through the lower surface of the body portion **2120**, and internal wirings **2135** electrically connecting the upper pads **2130** to the lower pads **2125** in the body portion **2120**. The upper pads **2130** may be electrically connected to connection structures **2400** (see FIG. **15**). The lower pads **2125** may be connected to wiring patterns **2005** on the principal substrate **2001** of the electronic system **2000** illustrated in FIG. **15** via conductive connection portions **2800**.

(135) Each of the semiconductor chips **2200** may include a semiconductor substrate **3010** and a first structure **3100** and a second structure **3200** sequentially stacked on the semiconductor substrate **3010**. The first structure **3100** may include a peripheral circuit region including a peripheral wirings **3110**. The first structure **3100** may include the peripheral circuit transistor **60TR** as described with reference to FIG. **8**. While the first structure **3100** is illustrated as having a same structure as a peripheral circuit structure of the integrated circuit device **200** illustrated in FIG. **8**, the inventive concept is not limited thereto.

(136) The second structure **3200** may include a common source line **3205**, a gate stack **3210** on the common source line **3205**, a channel structure **3220** passing through the gate stack **3210**, and a bit line **3240** electrically connected to the channel structure **3220**. The gate stack **3210** may include the first and second gate stacks **GS1** and **GS2** illustrated in FIG. **9**. The first and second gate stacks **GS1** and **GS2** may include gate electrodes **130**. Also, each of the semiconductor chips **2200** may include contact plug **CNT** electrically connected to the gate electrodes **130**.

(137) Each of the semiconductor chips **2200** may be electrically connected to the peripheral wirings **3110** of the first structure **3100**, and may include a through wiring **3245** extending into the second structure **3200**. The through wiring **3245** may be arranged outside the gate stack **3210**. In other embodiments, the semiconductor package **3003** may further include a through wiring passing through the gate stack **3210**. Each of the semiconductor chips **2200** may further include an I/O pad **2201** (see FIG. **15**) electrically connected to the peripheral wirings **3110** of the first structure **3100**.

(138) Referring to FIG. **17**, a semiconductor package **4003** has a similar configuration as the semiconductor package **3003** described with reference to FIG. **16**. However, the semiconductor package **4003** includes semiconductor chips **2200a**.

(139) Each of the semiconductor chips **2200a** may include a semiconductor substrate **4010**, a first structure **4100** on the semiconductor substrate **4010**, and a second structure **4200** bonded to the first

structure **4100** on the first structure **4100** by using a wafer bonding method.

(140) The first structure **4100** may include a peripheral circuit region including a peripheral wiring **4110** and first bonding structures **4150**. The first structure **4100** may include the peripheral circuit transistors **60TR** as described with reference to FIG. **8**. While the first structure **4100** is illustrated as having a same structure as a peripheral circuit structure of the integrated circuit device **200** illustrated in FIG. **8**, the inventive concept is not limited thereto.

(141) The second structure **4200** may include a common source line **4205**, a gate stack **4210** between the common source line **4205** and the first structure **4100**, and a channel structure **4220** passing through the gate stack **4210**. The gate stack **4210** may include the first and second gate stacks **GS1** and **GS2** illustrated in FIG. **9**. The first and second gate stacks **GS1** and **GS2** may include a gate electrodes **130**. Also, each of the semiconductor chips **2200a** may include contact plugs **CNT** electrically connected to the gate electrodes **130**.

(142) Also, each of the semiconductor chips **2200a** may include second bonding structures **4250** respectively electrically connected to the gate electrodes **130** of the gate stack **4210**. For example, some of the second bonding structures **4250** may be configured to be connected to a bit line **4240** electrically connected to the channel structure **4220**. Others of the second bonding structures **4250** may be configured to be electrically connected to the gate electrodes **130** via the contact plugs **CNT**.

(143) The first bonding structures **4150** of the first structure **4100** and the second bonding structures **4250** of the second structure **4200** may be bonded to each other while contacting each other. Portions of the first bonding structures **4150** and the second bonding structures **4250** that are bonded to each other may include a metal such as copper (Cu), but are not limited thereto.

(144) While the inventive concept has been particularly shown and described with reference to embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. An integrated circuit device comprising: a semiconductor substrate having a cell region and a connection region adjacent to the cell region; a gate stack including gate electrodes and insulating layers alternately stacked and having a stair-stepped structure in the connection region covered by a cover insulating layer, wherein each one of the gate electrodes extends from the cell region to the connection region to terminate in a corresponding pad portion, and the gate electrodes include lower gate electrodes disposed in a lower portion of the stair-stepped structure, upper gate electrodes disposed in an upper portion of the stair-stepped structure, and middle gate electrodes disposed between the lower gate electrodes and the upper gate electrodes; and a contact plug disposed in the connection region and comprising: first contact plug portions extending through the cover insulating layer to respectively contact pad portions of the lower gate electrodes; second contact plug portions extending through the cover insulating layer to respectively contact pad portions of the middle gate electrodes; and third contact plug portions extending through the cover insulating layer to respectively contact pad portions of the upper gate electrodes, wherein a second diameter of at least one of the second contact plug portions at an uppermost surface of the at least one of the second contact plug portions is less than a first diameter of at least one of the first contact plug portions at an uppermost surface of the at least one of the first contact plug portions, and the second diameter is less than a third diameter of at least one of the third contact plug portions at an uppermost surface of the at least one of the third contact plug portions.
2. The integrated circuit device of claim 1, wherein the first diameter is greater than the second diameter, and the third diameter is greater than the second diameter and less than the first diameter.
3. The integrated circuit device of claim 2, wherein the second diameter ranges from about 90% to about 95% of the first diameter, and the second diameter ranges from about 95% to about 99% of

the third diameter.

4. The integrated circuit device of claim 2, wherein another one of the first contact plug portions has a diameter different from the first diameter, each one of the second contact plug portions has the second diameter, and each one of the third contact plug portions has the third diameter.
 5. The integrated circuit device of claim 4, wherein the at least one of the first contact plug portions contacts a pad portion of a lowermost gate electrode among the gate electrodes.
 6. The integrated circuit device of claim 1, wherein the second contact plug portions respectively contact pad portions of gate electrodes disposed between $\frac{1}{4}$ to $\frac{1}{2}$ of a height of the gate stack as measured from an upper surface of the semiconductor substrate.
 7. The integrated circuit device of claim 1, wherein a distance between the cell region and any one of the first contact plug portions is greater than a distance between the cell region and any one of the second contact plug portions, and a distance between the cell region and any one of the third contact plug portions is less than a distance between the cell region and any one of the second contact plug portions.
 8. The integrated circuit device of claim 1, wherein each one of the first contact plug portions, the second contact plug portions and the third contact plug portions has a tapered pillar shape.
 9. The integrated circuit device of claim 1, wherein, for each gate electrode of the gate electrodes, an upper surface of a respective pad portion of the gate electrode is a highest portion of the gate electrode.
 10. The integrated circuit device of claim 1, further comprising: a peripheral circuit structure disposed between the semiconductor substrate and the gate stack.
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